



Creating a CITATION.cff File

Making your software citable

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2026-09-23

A catalog record for your software

Nobody emails a book's author to ask how to cite it. The catalog record answers that.

Software has never had one, so people improvise: they cite a URL, or the paper the code appeared in, or nothing.

A `CITATION.cff` file is **the catalog record for a repository**: who to credit and how. Once it exists, GitHub adds a “Cite this repository” button.

What goes in a CITATION.cff?

Minimal file, today

- `title`
- `authors` (ORCID if you have one)
- `message`
- `version` (optional for now)

As the project grows

- release versions
- a DOI from Zenodo
- keywords and an abstract
- repository URLs

It works **before** any release or DOI. Update it when you have them.

If you've edited a repository record or a Dublin Core field, you've done harder metadata work than this. Zenodo and Zotero both read the file, and it travels with every clone and fork.

Create the file on GitHub

In your fork of `software-demo` :

1. **Add file** → **Create new file**
2. Name it exactly `CITATION.cff` , in the repo root (case-sensitive)
3. Click **Insert example** in the banner GitHub shows
4. Replace the placeholder names, add ORCIDs, set `title` and `url`
5. Delete the `doi` line. No release yet? Delete `version` and `date-released` too
6. **Commit changes...** to `main`

Prefer a form? [cffinit](#) walks you field by field and validates as you type. Paste its output into the same file.

Name it, insert the example, edit, commit

Adding a CITATION.cff file helps users to easily cite your software from the repository overview. [Learn more about CITATION files.](#)

2 Click Insert example → Insert example

research-software-citable-discoverable / CITATION.cff ← 1 Name it CITATION.cff

Cancel changes Commit changes...

4 Commit to main No wrap

```

1  cff-version: 1.2.0
2  message: "If you use this software, please cite it as below."
3  authors:
4    - family-names: "YOUR_NAME_HERE"
5      given-names: "YOUR_NAME_HERE"
6      orcid: "https://orcid.org/0000-0000-0000-0000"
7    - family-names: "Lisa"
8      given-names: "Mona"
9      orcid: "https://orcid.org/0000-0000-0000-0000"
10 title: "research-software-citable-discoverable"
11 version: 1.0.0
12 doi: 10.5281/zenodo.1234
13 date-released: 2026-07-05
14 url: "https://github.com/UC-OSPO-Network/research-software-citable-discoverable"
15

```

3 Edit the placeholders

Use `Control + Shift + m` to toggle the `tab` key moving focus. Alternatively, use `esc` then `tab` to move to the next interactive element on the page.

What a finished file looks like

```
1 cff-version: 1.2.0
2 title: "Biodiversity Analysis Toolkit"
3 message: "If you use this software,
4   please cite it as below."
5 authors:
6   - family-names: "Dennis"
7     given-names: "Tim"
8     orcid: "https://orcid.org/0000-0001-6632-3812"
9   - family-names: "Phan"
10    given-names: "Leigh"
11 version: "0.1.0"
12 date-released: 2026-02-01
13 url: "https://github.com/UC-OSPO-Network/software-demo"
```

No **doi** yet, on purpose. You mint one in Reid's episode and come back to add it.

Commit to the **default branch** and the Cite button appears.

Check your work against the [03-add-citation](#) branch.

Predict: what will GitHub do?

You're about to commit `CITATION.cff` to the default branch.

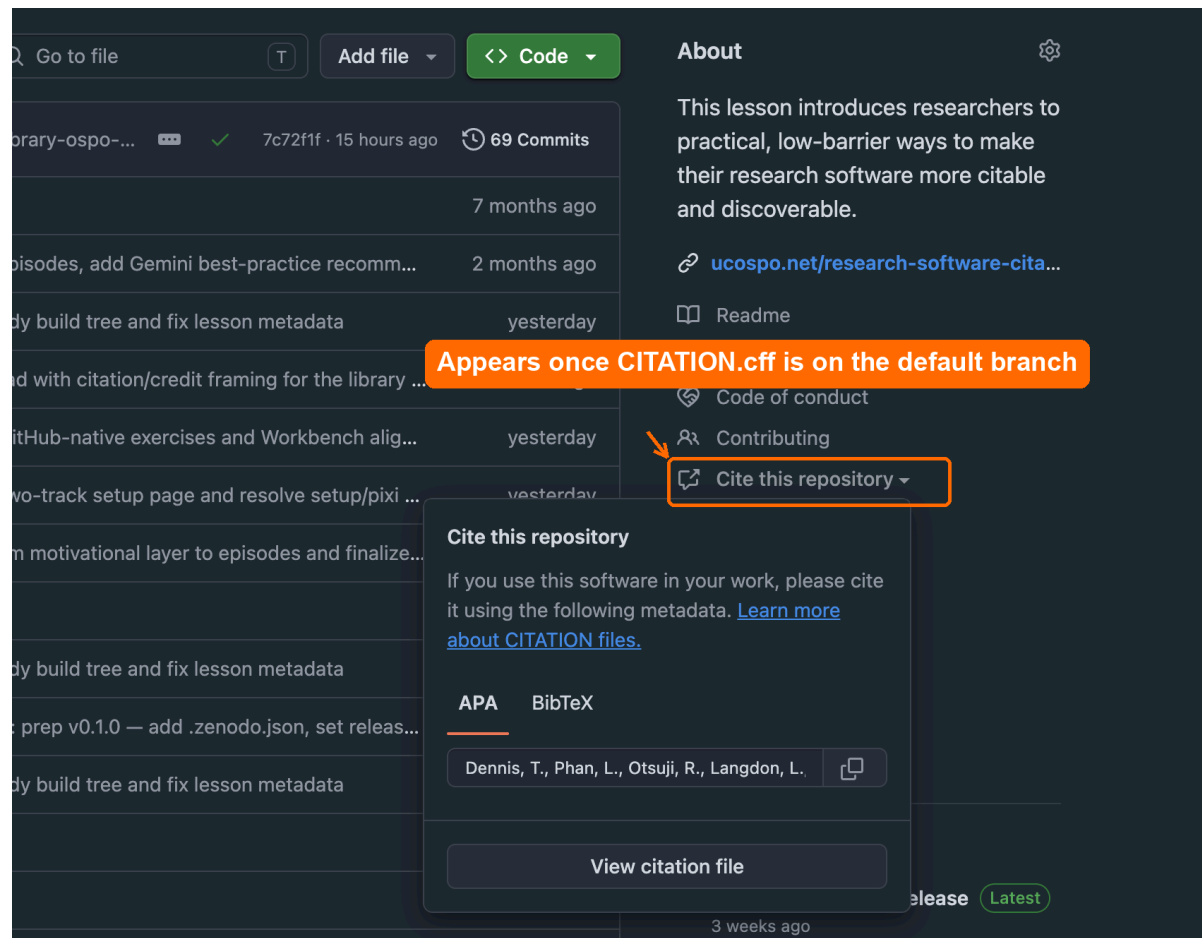
What changes on the repository's front page?

GitHub parses the file and adds a “**Cite this repository**” link to the About sidebar, with ready-made APA and BibTeX. No DOI required.

Cite this repository

Once the file is on the default branch, the About sidebar gets a **Cite this repository** link: APA, BibTeX, and a download of the CFF.

One gap: the generated citation leaves out `version`, even when the file has it. Tell people to add it by hand, or keep a plain citation in your README.



Pointing people to a paper

Want people to cite a journal article alongside, or instead of, the software? Add `preferred-citation` :

```
1 preferred-citation:
2   type: article
3   title: "Biodiversity Analysis at Scale: Methods and Software"
4   authors:
5     - family-names: "Dennis"
6       given-names: "Tim"
7   journal: "Journal of Open Source Software"
8   year: 2025
9   doi: "10.21105/joss.00000"
```

GitHub and Zotero then show the paper first. No paper yet? Leave it out; the rest of the file works without it.

Challenge: the button isn't showing

 2 min

A researcher added a `CITATION.cff`, but GitHub still shows no “Cite this repository” link. **What do you check?**

1. **Filename and location:** exactly `CITATION.cff`, in the root, not a subfolder
2. **Branch:** GitHub only reads it from the default branch
3. **Valid YAML:** one bad indent or missing quote stops the panel. cffinit catches these

Supporting others

In a consultation you rarely edit a researcher's files yourself. Make it easy for them to own:

- **Point them to cffinit** instead of hand-edited YAML. It validates as they go.
- **Show the payoff first:** open a repo that already has the Cite panel. People adopt what they can see.
- **Know the boundary:** if authorship or author order is contested, that's a credit negotiation between collaborators. Surface it, don't arbitrate it.

Challenge: add your CITATION.cff

 5 min

In your fork of `software-demo` :

1. Create `CITATION.cff` in the repo root (Insert example, or cffinit)
2. Add at least `title` , your name under `authors` , and `message` . Add your ORCID if you have one
3. Commit to `main` and refresh: look for **Cite this repository**

While you work: what metadata could you add today, and what would need a collaborator's input?

Challenge: is a citation file enough?

1. You change the code tomorrow. Does today's citation still point to the code they used?
2. The repo is renamed, or GitHub goes away. Does the citation still resolve?
3. Can a reader get the **exact** version behind a paper's results?



2 min

Not yet. `CITATION.cff` says **how** to cite the software. A tagged release archived to Zenodo gives the citation something durable **to point at**: a DOI for an immutable snapshot.

Key points

- `CITATION.cff` is the foundation of software citation
- You can add it before releases, DOIs, or version tags
- GitHub shows ready-made citations once it's on the default branch
- Start simple and expand as the project develops
- Necessary but not sufficient: a release and DOI make the citation durable

Up next

Reid Otsuji: Minting a DOI and Creating a Release

Then back to Tim for Improving Metadata and Discoverability.